

Algebra 1
Unit 4
Lesson 6 Practice

Name _____
Date _____
Period _____

For each real-world representation of a context write an inequality that represents the constraint.

1. A person has to be at least 18 years old to vote.
2. A job offers a salary between \$42,000 and \$50,000.
3. The highest altitude for skydiving without oxygen is 14,000 feet.

An automatic soap dispenser holds 18 ounces (oz) of soap. Each time someone places their hand under the soap dispenser releases 0.15 oz of soap. The function $f(x) = 18 - 0.15x$ models the amount of soap, in ounces, in the dispenser for x dispenses.

4. Find the value of x for which $f(x) = 13.2$.
5. In terms of the context, what does the value of x that is the solution to $f(x) = 13.2$ mean?
6. Find the x -intercept.
7. What does the x -intercept mean within the context?

8. What key feature is $f(0)$?

9. Select all of the constraints for the context.

- ☐ $x \geq 0$
- ☐ $x \leq 18$
- ☐ $x \leq 120$
- ☐ $f(x) \geq 0$
- ☐ $f(x) \leq 18$
- ☐ $f(x) \leq 120$

10. Graph the function in terms of the context.

